

rReverseDigits

Write a **recursive** C function `rReverseDigits()` that takes a non-negative integer argument `num` and returns an integer whose digits are obtained by reversing those of the argument number. The result is passed to the calling function through a pointer variable `result`. For example, if `num` is 1234, then the function should return 4321 through the pointer variable. If `num` is 10, then the function should return 1. You may assume that the last digit of the input number is not 0, i.e. the number will not be in the form of 1110, 1200, etc. The function prototype is given below:

A sample program template is given below to test the functions:

```
#include <stdio.h>
void rReverseDigits(int num, int *result);
int main()
{
    int result=0, number;

    printf("Enter a number: \n");
    scanf("%d", &number);
    rReverseDigits(number, &result);
    printf("rReverseDigits(): %d\n", result);
    return 0;
}
void rReverseDigits(int num, int *result)
{
    /* Write your code here */
}
```

Some sample input and output sessions are given below:

- (1) Test Case 1
Enter a number:
1234
rReverseDigits(): 4321
- (2) Test Case 2
Enter a number:
102
rReverseDigits(): 201
- (3) Test Case 3
Enter a number:
12934
rReverseDigits(): 43921
- (4) Test Case 4
Enter a number:
8
rReverseDigits(): 8

